

Full Board Discovery by the Knight: the Closed Tour and the Parity

Theorem monograph T07 of the chess-dynamics-lab program

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Keywords: knight's tour, Hamiltonian cycle, bipartiteness, discovery

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Setting

The knight's tour is the classical problem: the knight-particle must visit every square of the board exactly once. In our terminology this is the limiting case of the discovery monovariant (T06): a particle~“fully discovers” the board when its trajectory covers all 64 squares. A closed tour — a tour whose last square is joined to the first by a knight move — turns the trajectory into a cycle: the particle can repeat the discovery forever.

The task of this monograph is to prove the parity obstruction (a closed tour has even length), to present an explicit certificate of a closed tour on the 8×8 board, and to justify the construction method (the Warnsdorff rule with a deterministic tie-breaking). The certificate is frozen in results/knight_tour.json and verified by the protocol C10 of the polyglot core in all seven languages.

Theorem

Theorem T07 (full discovery by the knight). (i) The knight graph is bipartite by the square colour, so every closed tour has the even length $64 = 32 + 32$: a closed tour exists only for an even number of squares. (ii) On the 8×8 board a closed tour exists; an explicit certificate is the 64-square sequence $f5, h4, g2, \dots, e4, d6$ closed by the move $d6 \rightarrow f5$. (iii) The certificate is verified by 64 adjacency checks: all squares distinct, every consecutive pair and the pair (last, first) joined by a knight move.

Proof

(i) Parity. Every knight move changes the coordinate sum $f + r$ by the odd quantity $(\pm 1) + (\pm 2)$, so the square colour alternates along any trajectory: the knight graph is

bipartite with the parts being the colour classes of 32 vertices each (bipartiteness proved in T02). A closed cycle in a bipartite graph has even length: leaving a vertex and returning to it, the cycle makes an equal number of steps in each part. A full tour covers all 64 vertices, and a closed full tour has the length exactly 64 — an even number consistent with the parity $32 + 32$. On a board of odd side (an odd number of squares) the parts are unequal, and a closed tour is impossible — the parity obstruction.

(ii) Existence. The existence of a closed tour on the 8×8 board is a classical result (the board satisfies the conditions of Schwenk’s theorem on boards admitting a closed tour; historically the first example is due to Euler, 1759). We strengthen it with a constructive certificate: the 64-square sequence produced by the Warnsdorff rule (“move to the square with the fewest onward continuations”) with the deterministic tie-breaking by the minimal square index, starting at f5. The algorithm is deterministic, so the certificate is reproducible; it is frozen in the repository and is not recomputed.

(iii) Certificate verification. The verification requires no search: the 64 squares of the certificate are pairwise distinct (the uniqueness check), and every consecutive pair — including the closing pair (last, first) — satisfies the knight-move criterion: the coordinate differences in absolute value equal (1, 2). These are 64 adjacency checks, executed in any of the seven implementations of the polyglot core in milliseconds. A verified certificate is a constructive existence proof: a Hamiltonian cycle of the knight graph is presented explicitly. The theorem is proved.

Protocol data

The tour certificate (check C10)

Parameter	Value
Certificate start	f5
Closing move	d6 → f5
Length	64 squares, 64 knight steps (closed)
Square uniqueness	all 64 distinct
Construction method	Warnsdorff + deterministic tie-break
Adjacency checks	64
Certificate file	results/knight_tour.json

Summary of what is proved

We proved the parity obstruction for closed tours (bipartiteness implies evenness), presented and verified an explicit closed tour of the 8×8 board — a Hamiltonian cycle of the knight graph — and fixed a deterministic construction method. The knight-particle fully discovers the board in 64 steps and returns to the start: the discovery monovariant reaches the maximum 64 and the cycle closes — the limiting case of the flow picture of T06.

Verification

Check: the C10 line of the polyglot core (python3 polyglot/python/chess_core.py and analogues) — the certificate is read from the core constant and verified by 64 adjacency checks in all seven languages; in the laboratory the certificate is additionally cross-checked with the file results/knight_tour.json.

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